

Intercellular communication

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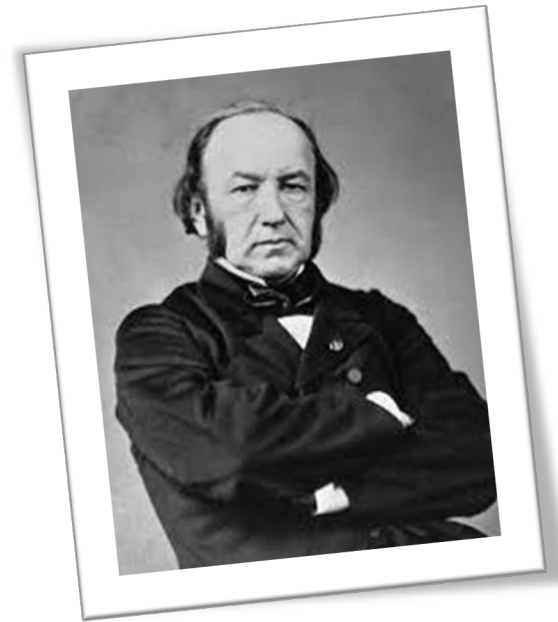
Definition of physiology

- Human physiology is the branch of biological sciences that studies the normal functioning of the human body — from the cell to the major systems (respiratory, cardiovascular, nervous, endocrine, etc.).
- Human physiology is in close relationship with medicine.
- Physiology also studies the interactions of an organism and its environment..
- .Study of the functions which ensure the maintenance of life of living beings.
- Human physiology studies the functional mechanisms which allow the human body to live, adapt and survive.

- Anatomy describes the structure, physiology explains how it works.
- It search to understand:
 - How cells produce energy
 - How the Heart Pumps Blood
 - How the lungs ensure gas exchange
 - How the brain coordinates functions
 - How the body maintains its internal balance

(homeostasis)

Claude Bernard



Definition of homeostasis

- **Tendency of organisms to maintain a relative internal stability.**
- **For an organism to be independent of the variations of the external environment, it must maintain stable the internal constants (temperature, pH, glycemia, pressure, ionic composition, etc.). (Claude Bernard 1872)**
- **The relative stability is maintained through compensatory or regulation mechanisms.**
- **Regulation can only be ensured through the communication between the different regulation systems.**

Intercellular communication

- The life of multicellular organisms is based on:
 - Communication
- Information exchange between cells

These interactions or exchanges control the functioning of the organs while determining homeostasis at the of the body

- **An essential condition for maintaining the life of cells:**
 - **cell-to-cell communication systems**

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- **How do the cells communicate with each other or What are the modes of communication?**

- Modes of intercellular communication:
- Exchanges and membrane transport
- Electric Transmission
- The Interaction Receptor Ligands

Support: The plasma membrane

Examples:

- Transmitting nervous messages, ensuring a relationship between organs, being informed about the environment, developing behaviours:
 - This is the role of the nervous system
- Controlling, monitoring the growth of certain tissues, regulating the production of substances necessary for the body: this is the mission of the hormonal system

Cellular
communication
modalities

**Endocrine
Communication**

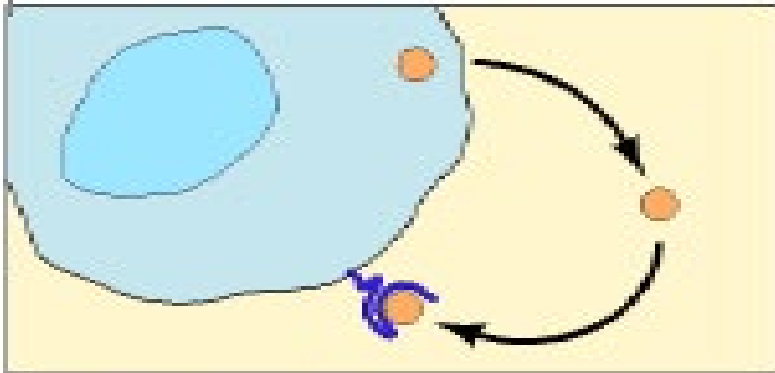
**Paracrine
Communication**

**Autocrine
Communication**

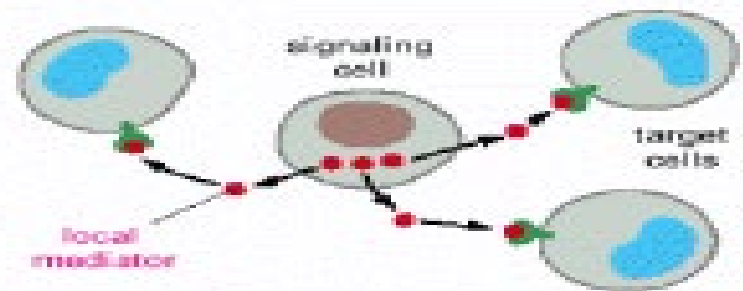
**Juxtacrine
Communication**

**Synaptic
Communication**

(A) AUTOCRINE



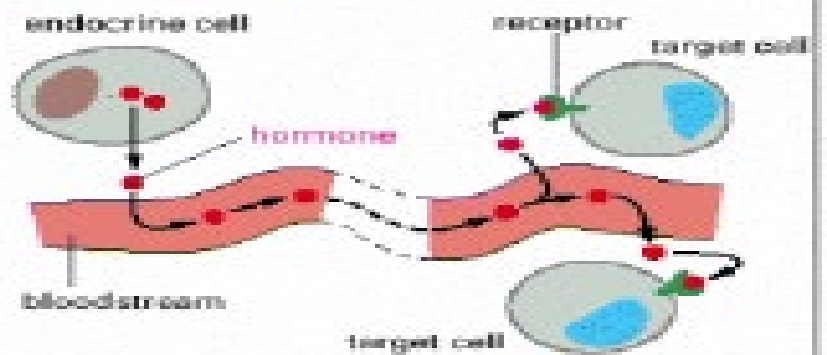
(B) PARACRINE



(D) SYNAPTIC



(C) ENDOCRINE



Intercellular communication by signaling molecules or ligands

- Signaling molecules are chemical substances from cells, able to play a messenger role by putting in communication 2 cells more or less far apart.
- They are also called: The ligands.
- Belong to several biochemical families or substances.

- Signaling molecules or Ligands
- Neurotransmitters: e.g. adrenaline, acetylcholine...
- Hormones and neurohormones

- Cytokines
- Immunoglobulins
- Eicosanoids (Derivatives of arachidonic acid such as prostaglandins and leukotrienes)
- Gases: NO

➤ Different types of communications **Endocrine**
Communication:

- Allows connecting by chemical signals cells located at a distance from each other.
- The signaling molecules are hormones.
- Are also called: LIGAND.
- Use the blood circulation to reach the target cell.

➤ **Nature of signaling molecules:**

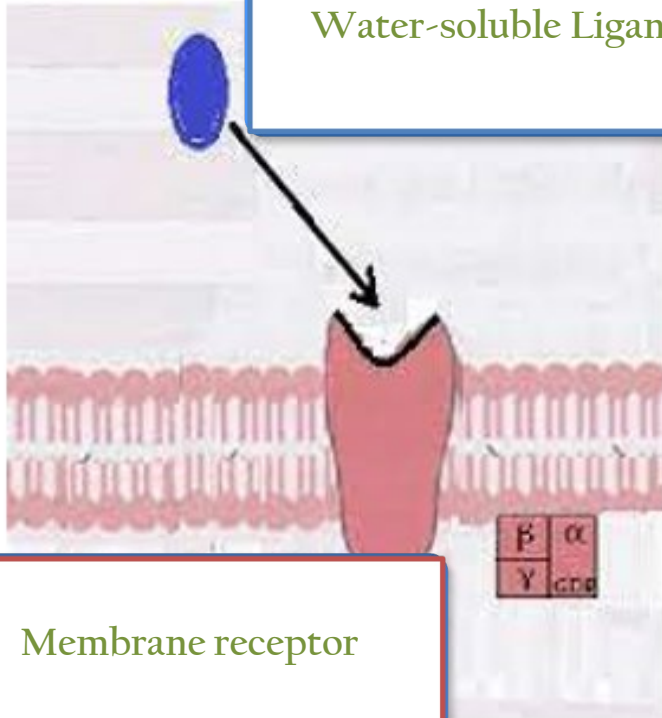
➤ **Water-soluble**

➤ **Lipid-soluble**

➤ Water-soluble molecules: High molecular weight, these are peptides, peptide derivatives or amino acid derivatives.

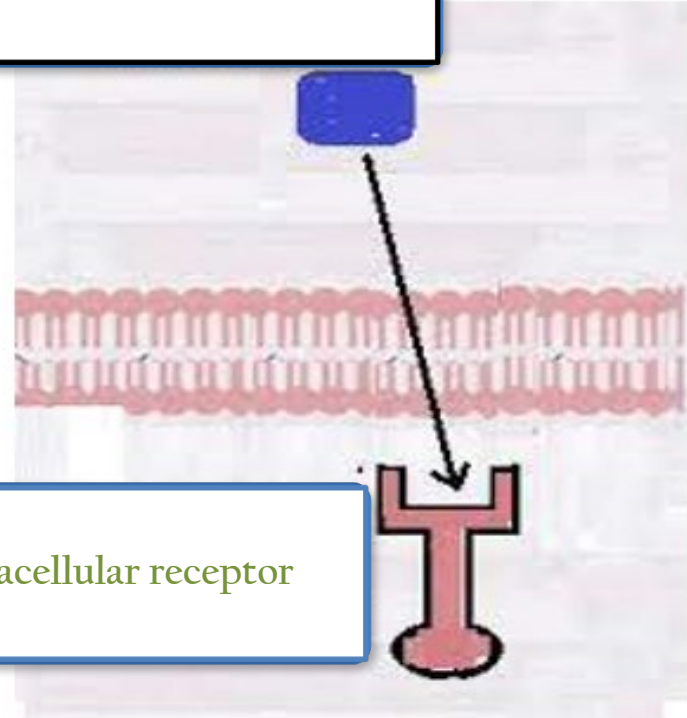
➤ The ligand-receptor interaction is a series of reactions leading to various modifications of cellular behavior.

Water-soluble Ligands



Membrane receptor

Lipid-soluble Ligands



intracellular receptor

The receptors: Membrane receptors

- Membrane proteins that have the capacity to:
 - Specifically recognize a signaling molecule or ligand
 - Induce modifications at the surface or inside the cell following the occupation of the receptor by the ligand
- Three classes of membrane receptors are distinguished:
 - Channel-receptors
 - Enzyme-receptors
 - G-protein coupled receptors (GPCRs)
- More than 50% of current therapeutic agents target GPCRs

Signaling: The phases of intercellular communication

Reception (The Capture):

The target cell detects the ligand via a **specific receptor** (usually on the surface, sometimes intracellular)

Transduction (The Conversion):

The binding triggers a **cascade of enzymatic reactions**.
This converts the initial extracellular signal into a meaningful internal message

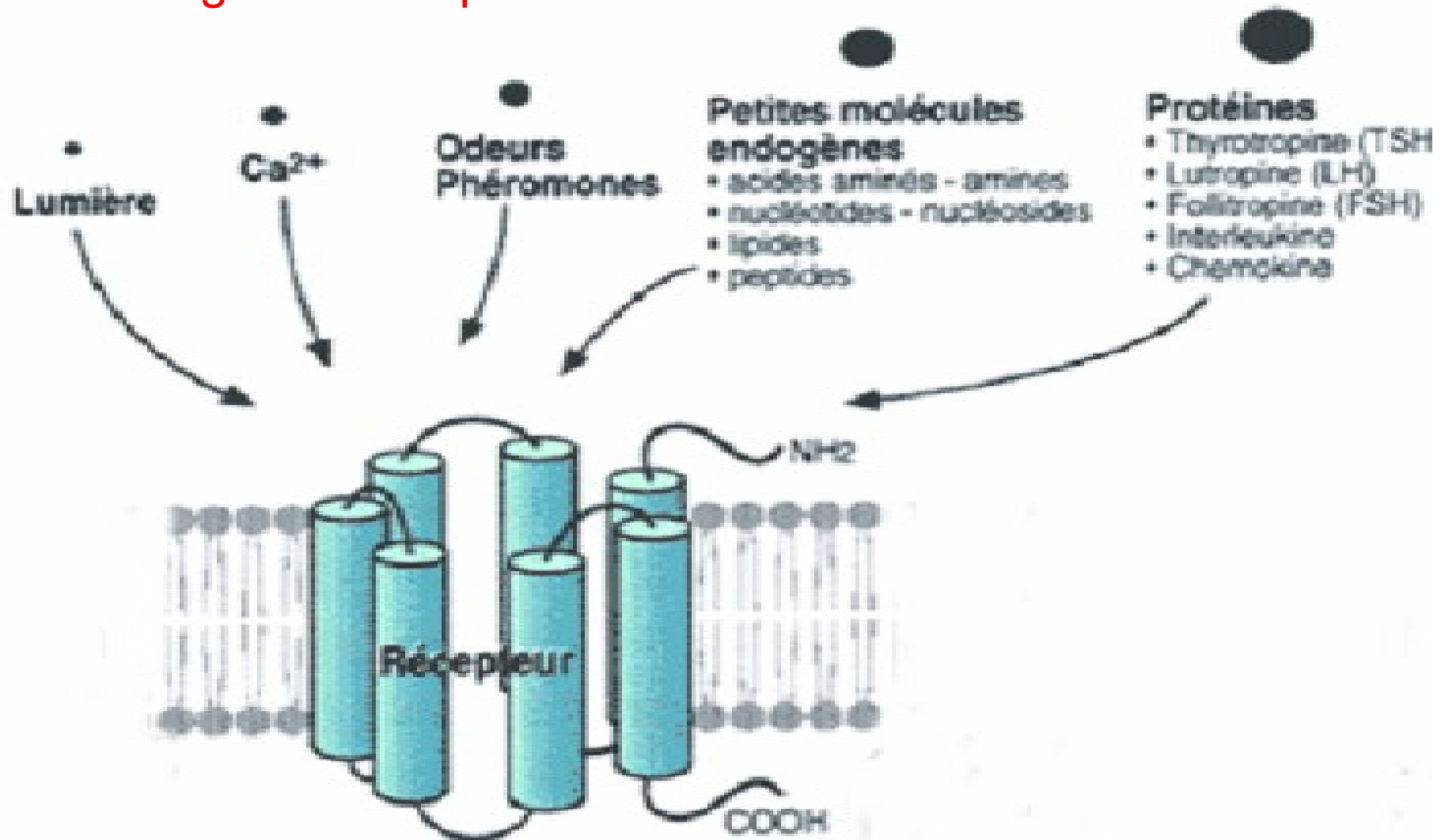
Response (The Action):

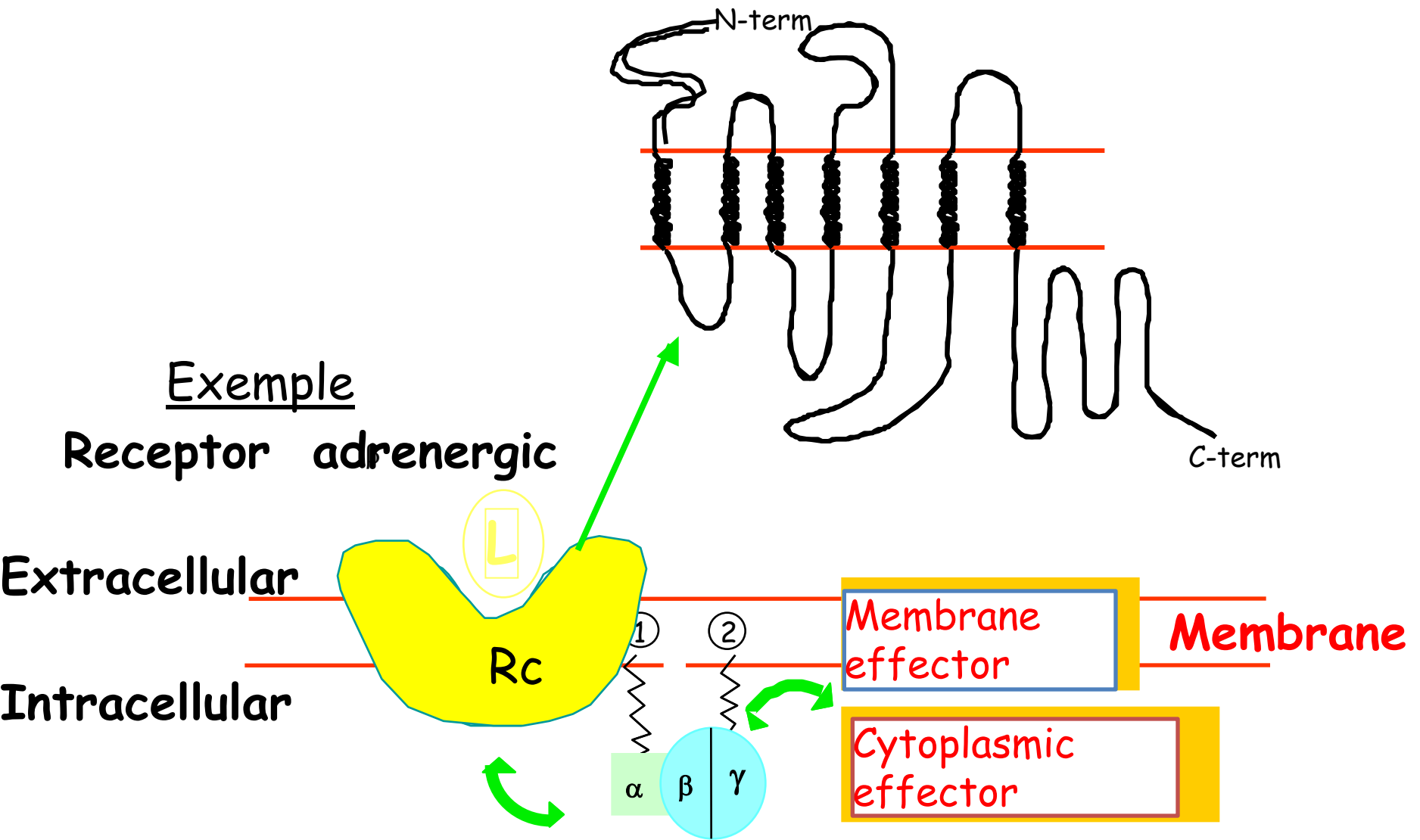
Which leads to the final response. who is the modification of cellular activity or function

Translates into a specific effect at the organism.

G-protein-coupled receptors

GPCR ligands: Ubiquitous





Nobel Prize in Physiology and Medicine 1994



Alfred G. Gilman



Martin Rodbell

Second messenger concept and signal amplification;

After the *GPCR* has been activated by its ligand.

Activation of an enzymatic effector which in turn activates a second Messenger in order to amplify the message

This second messenger also amplifies the signal and generates a massive cellular response

Concept of second messengers

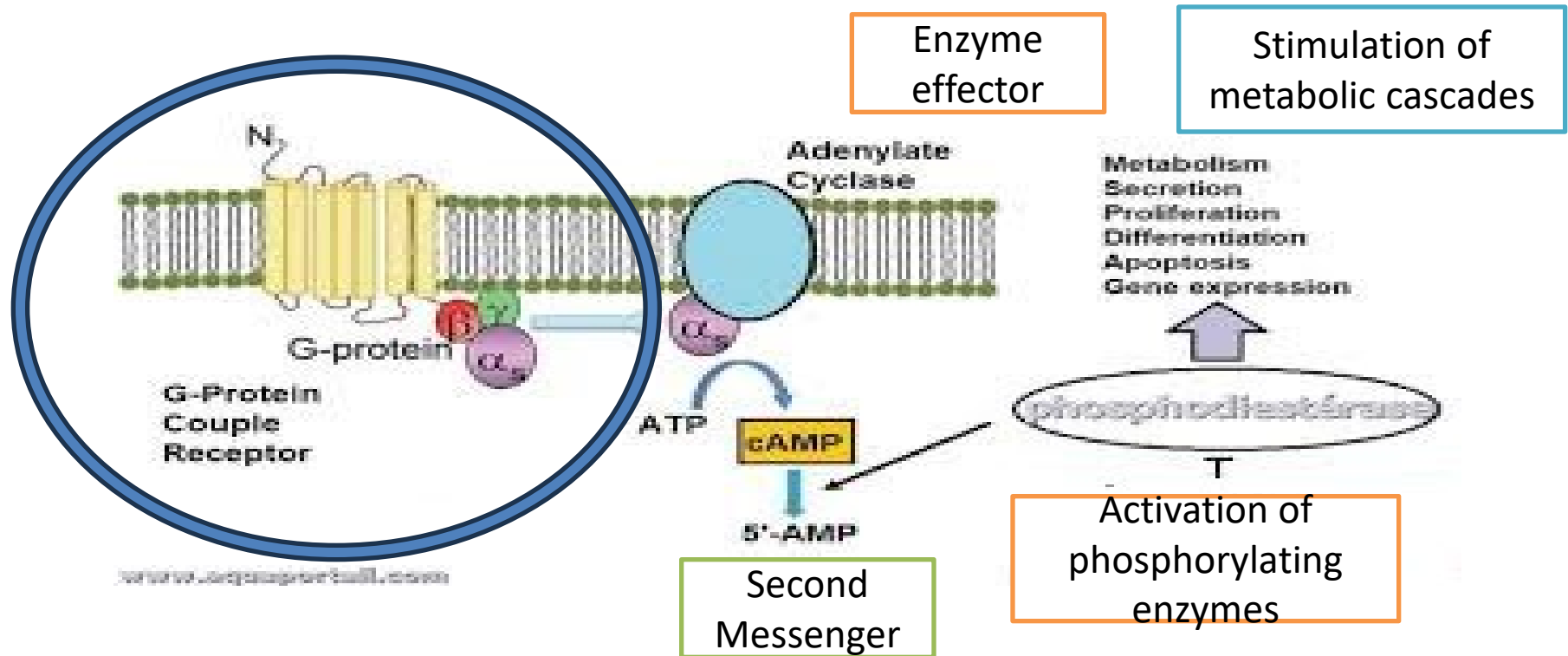
- AMPLIFIER: ADENYLATE CYCLASE
- Formation of cAMP: Second messenger
- cAMP occupies a Cytoplasmic Receptor
- ACTIVATES A PROTEIN KINASE: cAMP-Protein Kinase
- The complex phosphorylates a protein
- Phosphorylated protein  Cellular response

➤ G-protein-activated effectors:

- **The goal: Signal amplification**

- **Adenylate cyclase**: Synthesis of cAMP, the second messenger.
- **Phospholipases: C and D.**
- **Phospholipase C**: Catalyzes PIP₂ into IP₃ and diacylglycerol (DAG). These latter behave as second messengers.
- **G-protein can control ion channels**; in this case, the ion channel can be the effector protein of the G-protein

The cAMP (Cyclic Adenosine Monophosphate) Pathway

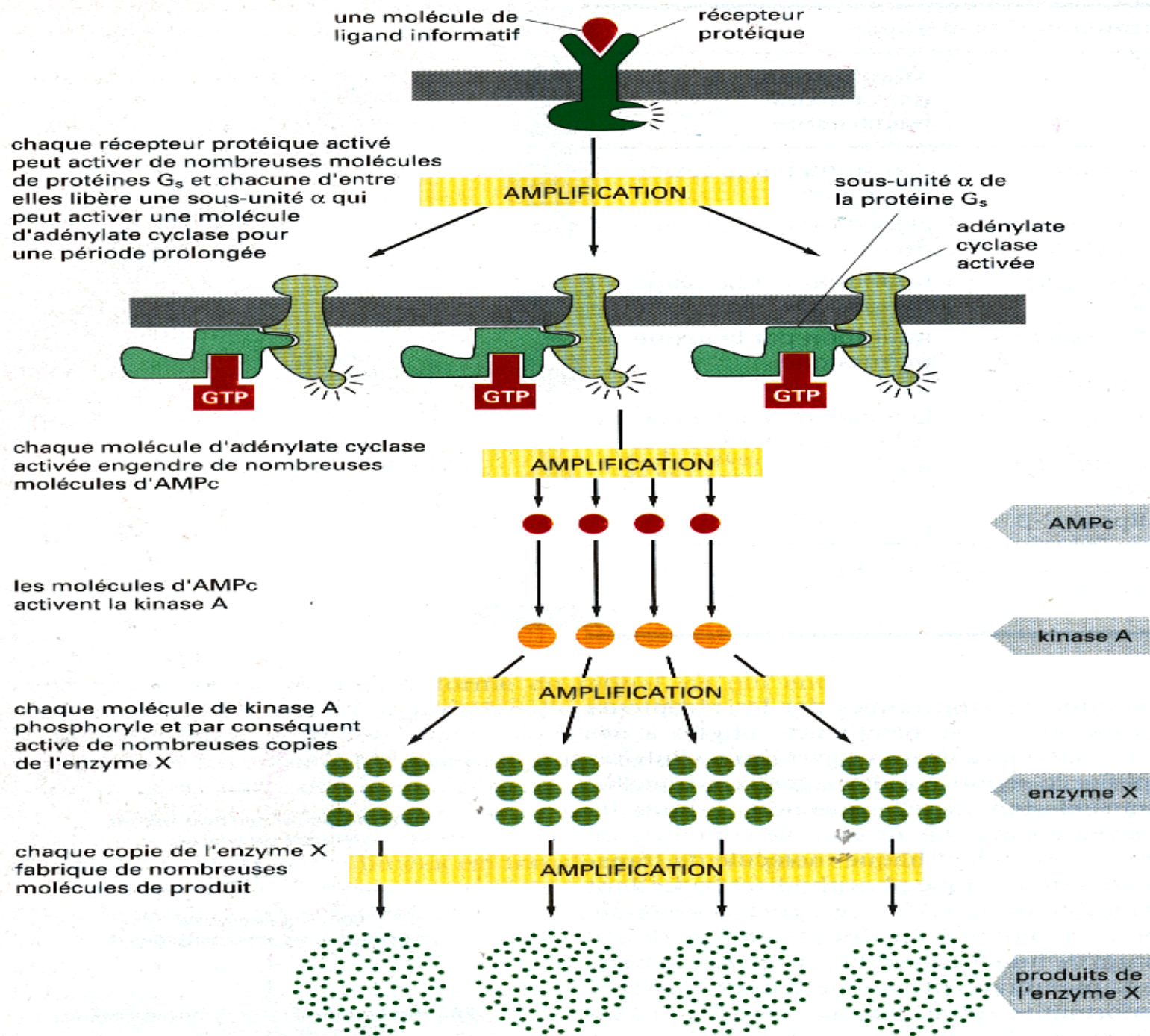


➤ **G-Protein Coupled Receptors (GPCR):**

- These are receptors linked to a G-protein; the latter are composed of three subunits: α (α_s for stimulation or α_i for inhibition), β , and γ .
- The α subunit possesses a binding site for GDP/GTP.

➤ **Transduction mechanism:**

- The message must pass through 7 steps, in order to be received and correctly interpreted by the target cell.
- The ligand binds to the GPCR that is specific to it;
- The receptor modifies its three-dimensional conformation and activates the G-protein;
- GDP leaves the α subunit of the G-protein (GP);
- GTP replaces the GDP on the nucleotide binding site of the α subunit;
- The α subunit is activated by binding with the GTP and detaches from the $\beta\gamma$ dimer of the G-protein

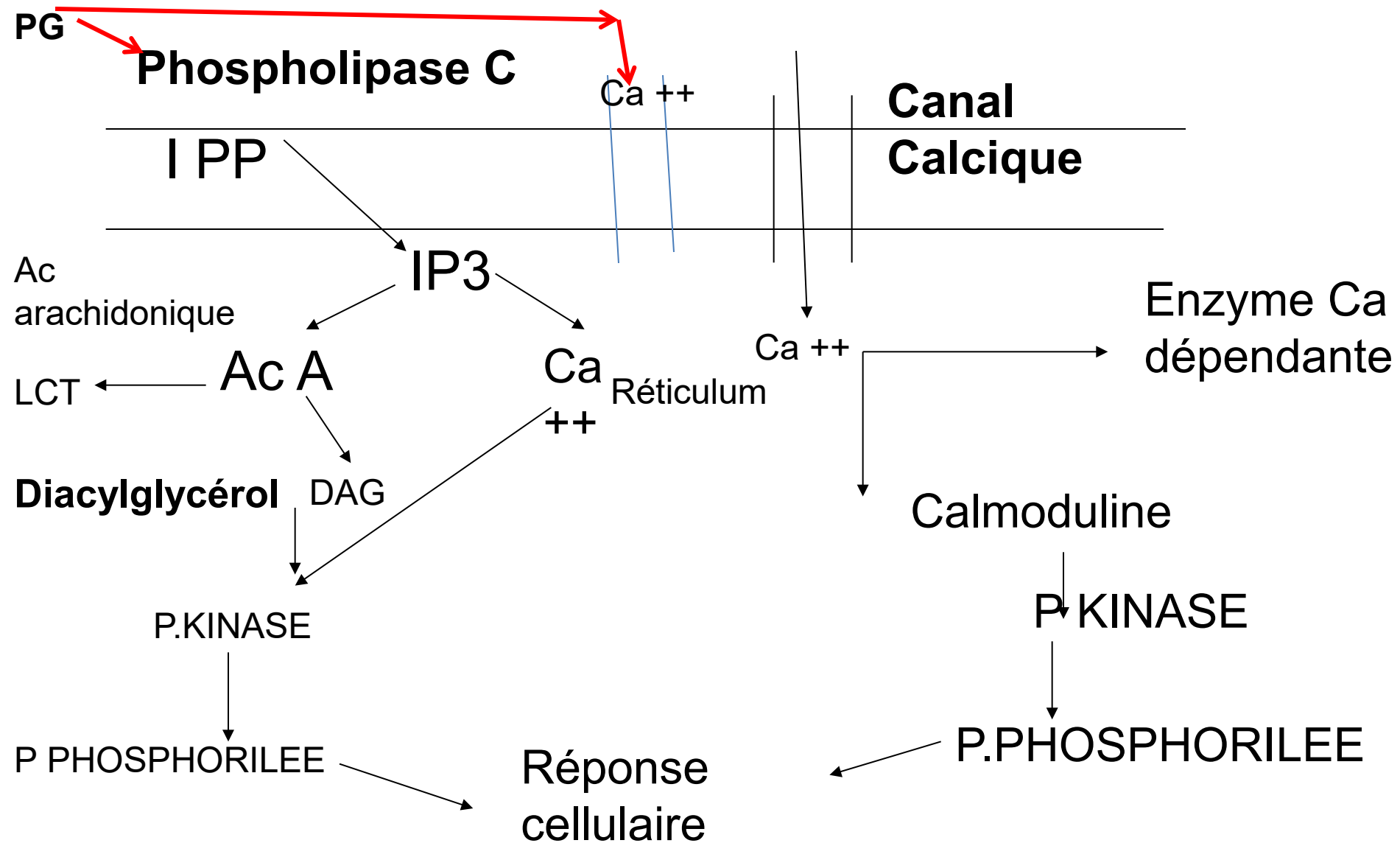


➤ **Clinical relevance of cell signaling dysfunctions:**

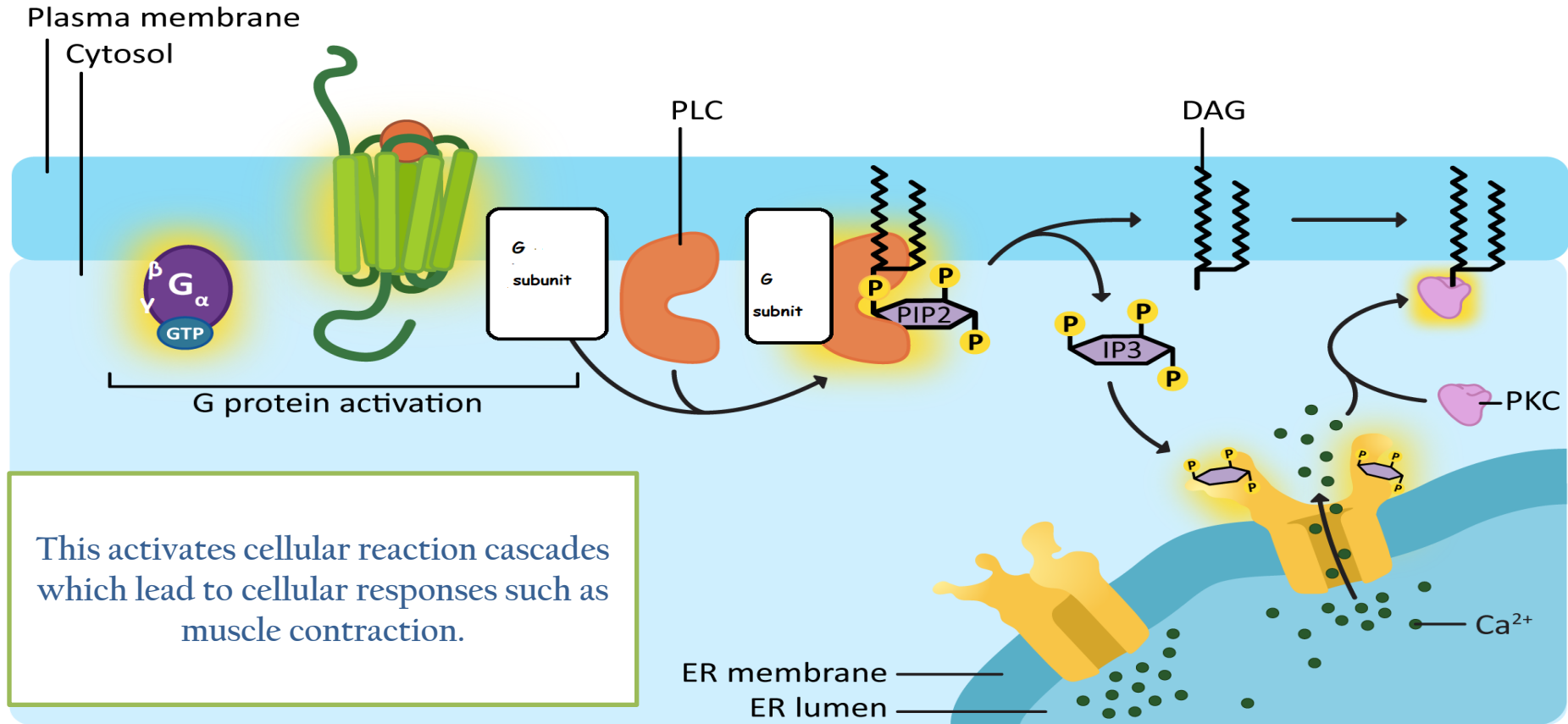
✓ **Cholera: Cholera toxin keeps adenylate cyclase permanently active in intestinal cells → continuous production of cAMP → massive diarrhea and severe dehydration.**

- **Certain genetic mutations of the α subunit are involved in the genesis of certain cancers:**
- **Pituitary gland tumours**
- **Ovarian tumors**

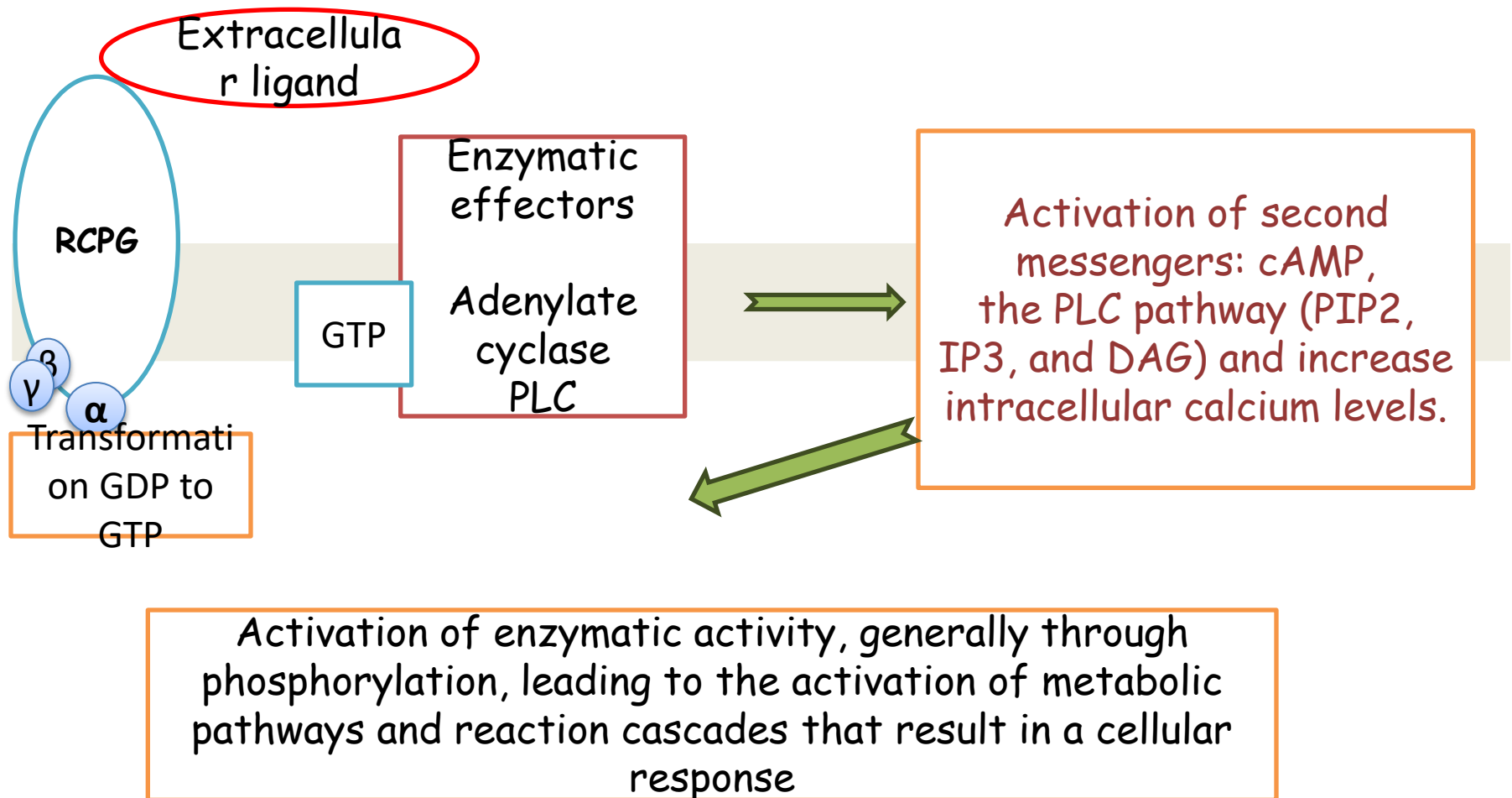
PG and ion channel coupled receptors, phospholipases and calcium



The IP₃ / DAG / Ca²⁺ Pathway



RCPG RECAP :



the cGMP (cyclic guanosine monophosphate) pathway

➤ Mechanism:

- Activation of the enzyme **guanylate cyclase**, which converts **GTP into cGMP**.
- **Action:** cGMP primarily acts by activating **Protein Kinase G (PKG)** and can also directly modulate the opening of specific **ion channels**

➤ Biological Effects:

-
- Vasodilation:
- Vision:

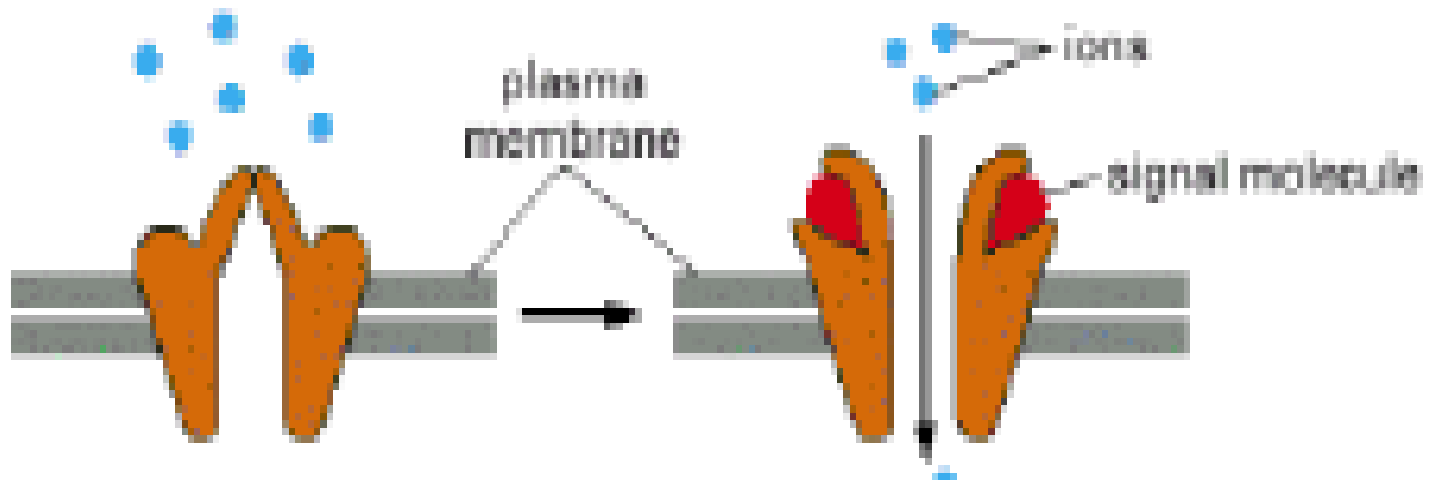
la voie GMPc

- Unlike cAMP, cGMP can be regulated by a G protein. In the visual system, the G protein (transducin) regulates cGMP.
- Furthermore, cGMP can be produced following activation by the enzyme guanylate cyclase, which is itself stimulated by the gas (NO).

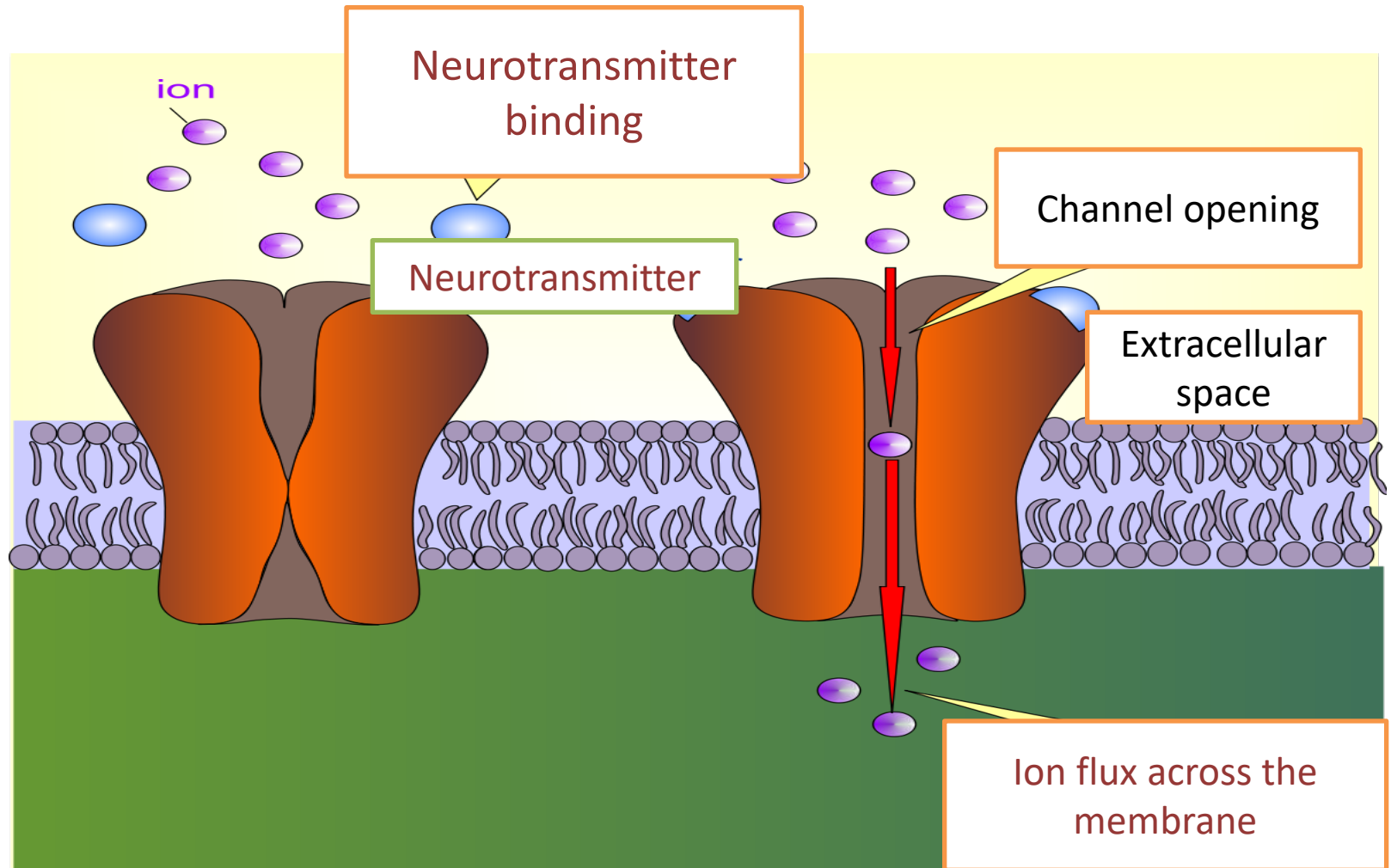
Ion channel-coupled receptors

Ligand-gated ion channels

A. Récepteurs canaux ioniques ligand-dépendant



Ion channel-coupled receptors



Enzyme linked receptor

- Enzyme-linked receptors are single-pass transmembrane proteins composed of two parts:
- An extrinsic (extracellular) part containing the ligand-binding site and an intrinsic (intracellular) part carrying the catalytic domain.
- The binding of the ligand to the extrinsic part stimulates the direct activation of the intrinsic enzymatic domain and the activation of cellular metabolic pathways.
- These receptors regulate cell growth, proliferation, differentiation, and survival, playing a crucial role in embryonic development and wound healing.

Different classes of enzyme receptors in relation to their specificity

- The receptor itself can have enzymatic activity
 - **Guanylate cyclase receptors**
 - cGMP production (Production de GMPc)
 - **Tyrosine phosphatase receptors**
 - Remove phosphate groups from phosphorylated tyrosines
 - **Tyrosine kinase receptors**
 - Phosphorylate tyrosine residues
 - **Serine/threonine kinase receptors**
 - Phosphorylate tyrosine residues

Enzyme-Linked Receptor ; Structure Characteristics and activation

.-Architecture:

- Extracellular (N-terminal): Contains the ligand-binding site.

- Intracellular (C-terminal): Carries the enzymatic domain.

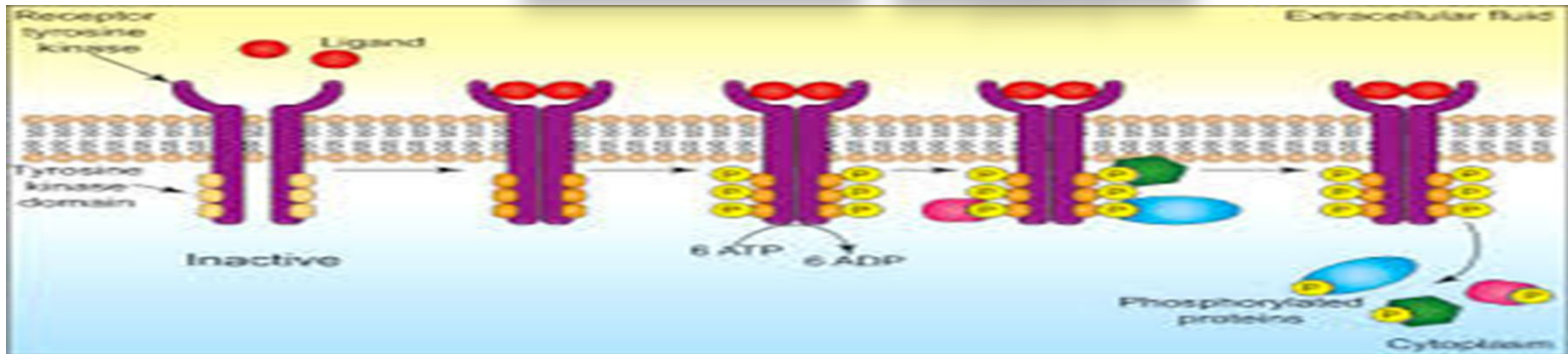
Ligand binding

Receptor dimerization

Generally an Autophosphorylation

Activation of signaling metabolic pathway

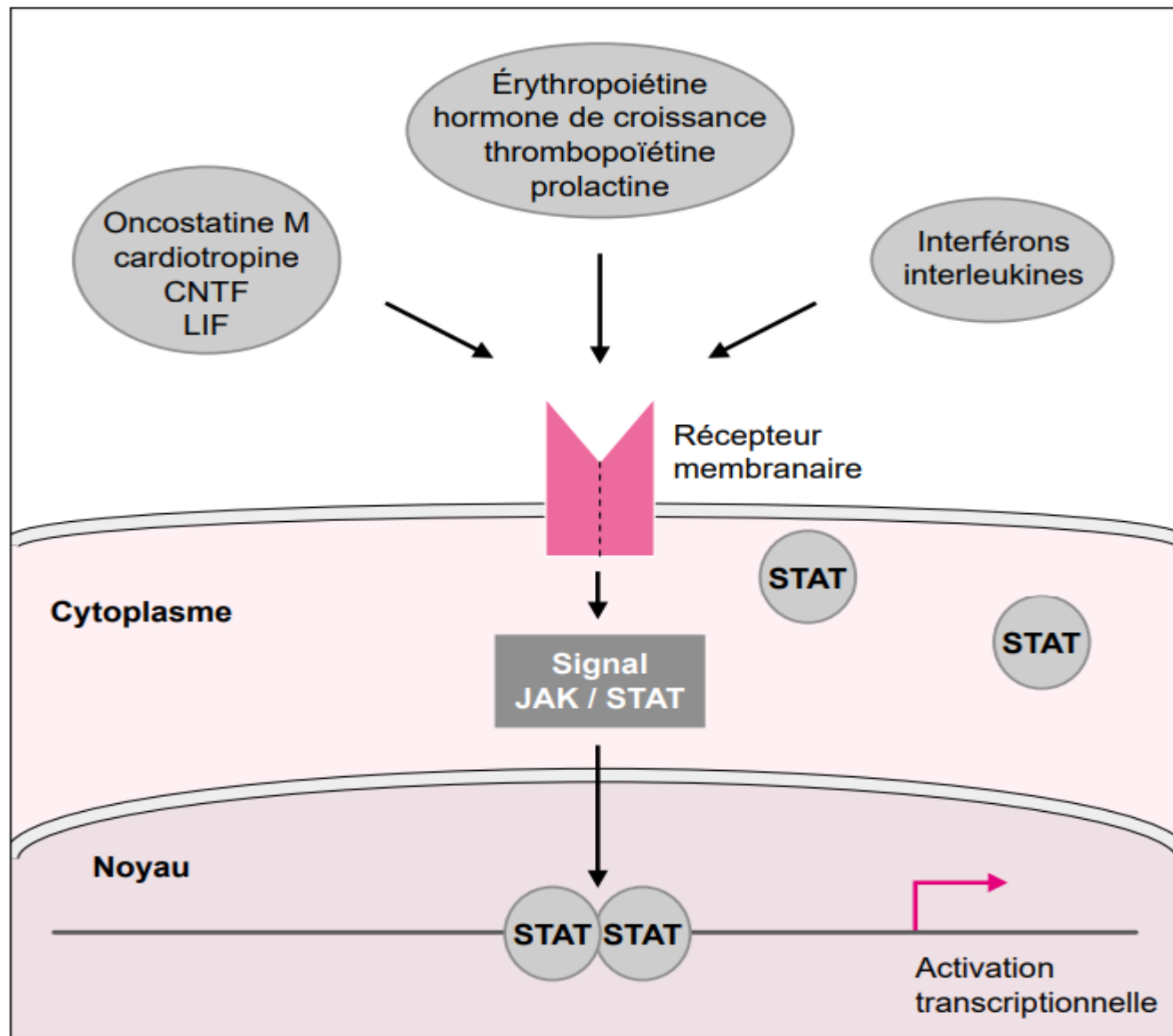
Cellular response



Special case of receptors that interact with JAK (Janus Kinase)

Special case of receptors interacting with JAK (Janus Kinase)

- The fields of application for JAK/STAT pathways are multiple, from cellular proliferation to immune function control.
- Understanding them allows for new therapeutic approaches, particularly in certain leukemias and hereditary immune deficiencies.
- The Janus Kinase (Jak) / STAT signaling cascade is a ubiquitous intracellular signaling pathway activated by several cytokines and growth factors.
- This pathway is responsible for the effect of cytokines on cellular proliferation, differentiation, survival, and host resistance to pathogens.
- Involved in several forms of cancer



Voies de transmission du signal JAK/STAT en réponse aux cytokines.

Currently treating systemic inflammatory diseases with anti-JAK

- Crohn's Disease, Ulcerative Colitis, and IBD (Inflammatory Bowel Disease)
- Tofacitinib: An anti-Janus Kinase (JAK2 and JAK3 inhibitor)
- Inhibition of intracellular signaling by blocking signaling pathways.
- Inhibition of STAT protein phosphorylation by JAKs, which blocks the production of inflammatory cytokines and TNF-alpha.

- Anti-fibrotics: nintedanib and pirfenidone
- Mechanism: Inhibit tyrosine kinase, thereby blocking the activity of the fibroblast growth factor receptor and the vascular endothelial growth factor receptor.
- They block intracellular signaling.

Second Messenger Interaction

AMPc

Ca⁺⁺

GMPc

Synarchic action

- cAMP controls calcium metabolism
- Ca⁺⁺ controls cAMP metabolism
- cAMP modulates the sensitivity to Ca⁺⁺ of elements involved in the Ca⁺⁺-dependent cellular response
- Ca⁺⁺ and cAMP control the same cellular functions (secretion, muscle contraction)

The synarchic functioning can be carried out in several ways:

- ✓ Coordinated operation
- ✓ Hierarchical functioning
- ✓ Redundant operation
- ✓ Antagonist functioning
- ✓ Sequential operation

Signal Termination: Definition and Mechanisms

Signal termination is the essential biological process that ends a cellular response after stimulation.

To prevent prolonged activation that could exhaust or damage the cell.

To allow the cell to become sensitive again and ready to receive a new message.

Message Interruption

➤ Catabolism of second messengers

- Ca^{2+} : Four mechanisms

Expulsion of Ca^{2+} to the exterior by a Ca^{2+} -dependent ATPase pump.

Reuptake of Ca^{2+} by the sarcoplasmic reticulum via an ATPase pump.

Reuptake by the mitochondria.

Sodium-calcium exchanger system that moves Ca^{2+} out into the extracellular medium.

- cAMP: Degraded by phosphodiesterase The latter (phosphodiesterase) is inhibited by theophylline and caffeine.

References

Physiologie Humaine : H Guénard 2019

Physiologie Humaine : Vander Maloine 2017